

Quarterly Operations Review

Q3 2026 Performance & Strategic Outlook

Supply Chain & Operations Analytics Team

September 21, 2026

Forecast Accuracy	Service Level	Cost Savings	Safety Index
92.4%	97.8%	\$847K	98.2%

Executive Summary

Bottom Line Up Front (BLUF)

Our supply chain optimization initiative has delivered **\$847K in annual savings** through improved demand forecasting and inventory management. Key achievements:

- **Forecast accuracy improved to 92.4%** (MAPE of 7.6%) using Prophet model with holiday/promotion regressors
- **Service level increased from 89% to 97.8%** through optimized safety stock policy
- **Distribution costs reduced by 15%** via linear programming optimization
- **Stockout incidents reduced by 78%** compared to previous quarter

Recommendation: Approve full deployment of optimized inventory policy across all 5 distribution centers.

Safety & Equipment Health

Safety Performance (Weeks 6-7 Recap)

Metric	Target	Actual	Status
TRIR (Total Recordable Incident Rate)	< 2.0	1.4	On Track
Near-Miss Reports	> 50/month	67	Exceeds
Safety Training Completion	100%	98.2%	On Track
Equipment Downtime	< 3%	2.1%	On Track
Predictive Maintenance Alerts	N/A	23 resolved	Active

Key Insights:

- Zero lost-time injuries for 127 consecutive days
- Predictive maintenance reduced unplanned downtime by 34%
- Forklift fleet health at 96% - scheduled Q4 maintenance for aging units

Equipment Health Dashboard

Critical Equipment Status

Equipment	Health Score	Next Service	Risk Level
Conveyor System A	94%	Oct 15	Low
Conveyor System B	87%	Sep 28	Medium
Packaging Line 1	96%	Nov 1	Low
Packaging Line 2	91%	Oct 8	Low
Forklift Fleet (12 units)	89%	Rolling	Medium
HVAC - Warehouse	82%	Sep 25	Medium

Action Items:

- Priority: HVAC maintenance scheduled for Sep 25 to prevent Q4 issues
- Conveyor B bearing replacement planned - parts on order
- Budget request: \$45K for forklift battery replacements (3 units)

Supply Chain: Demand Forecasting

Prophet Model Performance

Metric	Value	Industry Benchmark	Assessment
MAPE	7.6%	< 15%	Excellent
RMSE	48.2 units	N/A	Good
Forecast Horizon	60 days	30-90 days	Standard
External Factors	Holidays + Promotions	Varies	Enhanced

Key Findings:

- Weekly seasonality: 20% higher demand Mon-Fri vs weekends
- Q4 holiday uplift: 25-35% above baseline (Nov-Dec)
- Promotion events drive 25% demand spikes - now predictable
- Weather integration planned for Q4 (potential 3% accuracy gain)

Supply Chain: Inventory Optimization

Safety Stock & Reorder Point Analysis

Parameter	Old Policy	New Policy	Impact
Service Level Target	90%	95%	+5%
Safety Stock	892 units	1,247 units	+355 units
Reorder Point	4,392 units	4,747 units	+355 units
Stockout Days/Year	38 days	8 days	-79%
Annual Holding Cost	\$651K	\$912K	+\$261K
Annual Stockout Cost	\$1,425K	\$300K	-\$1,125K

NET ANNUAL SAVINGS: \$864,000

(Stockout cost reduction minus additional holding cost)

Supply Chain: Distribution Network

Linear Programming Optimization Results

Route	Volume (units/day)	Cost/Unit	Daily Cost
W1_Central -> Store_A	800	\$2.50	\$2,000
W2_North -> Store_B	650	\$2.00	\$1,300
W2_North -> Store_D	550	\$2.50	\$1,375
W3_South -> Store_C	900	\$2.00	\$1,800
W3_South -> Store_E	700	\$2.50	\$1,750
TOTAL	3,600	Avg: \$2.28	\$8,225

Optimization Impact:

- 15% cost reduction vs. previous equal-distribution policy
- Warehouse utilization balanced: 67% (W1), 80% (W2), 89% (W3)
- Sensitivity analysis: Network can absorb 18% demand surge before capacity constraints

Strategic Recommendations

1. Deploy Optimized Safety Stock Policy Enterprise-Wide

Investment	Expected ROI	Payback	Risk
\$261K (holding cost increase)	\$864K net savings/year	3.6 months	Low

2. Expand W2_North Warehouse Capacity by 500 Units

Investment	Expected ROI	Payback	Risk
\$125K (infrastructure)	\$180K/year (growth capacity)	8.3 months	Medium

3. Integrate Weather Data into Forecasting Model

Investment	Expected ROI	Payback	Risk
\$35K (API + development)	\$95K/year (accuracy gains)	4.4 months	Low

TOTAL PROJECTED ANNUAL VALUE: \$1.14M

Anticipated Questions

CFO Question: "Why do we need so much safety stock? That ties up working capital."

Response:

Great question. The safety stock increase of 355 units represents a \$261K increase in holding costs annually. However, our simulation data shows this investment prevents an estimated \$1.125M in stockout costs - that's a **4.3:1 return ratio**.

The math breaks down as follows:

- **Without safety stock:** 38 stockout days/year = \$1.425M lost (expediting + lost sales)
- **With optimized safety stock:** 8 stockout days/year = \$300K lost
- **Additional holding cost:** \$261K/year
- **Net benefit:** \$1.125M - \$261K = **\$864K annual savings**

Additionally, our 95% service level protects customer relationships. Each stockout event risks losing a customer permanently - the lifetime value impact isn't captured in these numbers but represents significant additional downside protection.

Working capital impact is offset by 3.6-month payback period.